Formal Lab Report - Sample A

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November 29, 2023

Chem 2081 Section 7

Introduction: In this lab an unknown was identified using nuclear magnetic resonance Spectroscopy, Fourier transform infrared spectroscopy, and mass spectrometry. Spectroscopy is important in determining the quality of known substances. It has many uses in wastewater analysis, pharmaceutical quality control and food analysis.

Procedure:

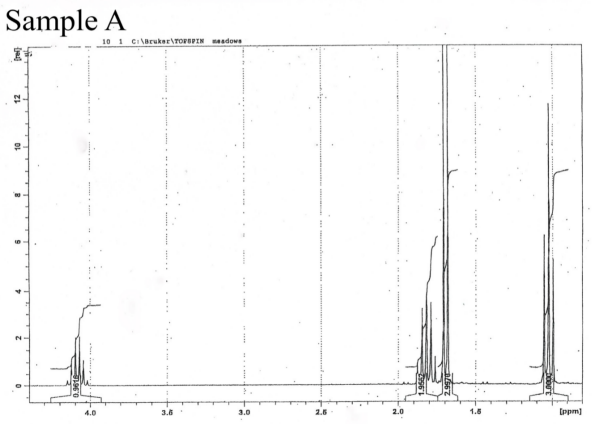
1. NMR or nuclear magnetic resonance is used to identify molecular structures (6). NMR is used to study nuclei based on their interaction with electromagnetic radiation. An NMR spectrometer uses an induced magnetic field to determine alpha or beta spin (5). The resonant frequencies make a graph called an NMR spectrum which will display peaks that are critical to the identification of the molecule (9). Integration is used to determine the number of protons present and the chemical shifts tell of the structure of the molecule. The peaks are indicative of the presence of protons and neighbors. Integration is used to count the protons. One peak shows a singlet, two peaks show a doublet, three a triplet, and four a quartet (4).

2. Fourier transform infrared spectroscopy or FTIR is the most common form of infrared spectroscopy. It does not destroy the sample, is quicker than other techniques, and provides a more accurate reading (7). An electromagnetic radiation excites all of the H and C at the same time. This makes a continuous oscillation. The oscillation curve is called a free-induction decay. FTIR can be used to show functional groups present in a molecule. This is read based on where peaks fall on the graph (4).

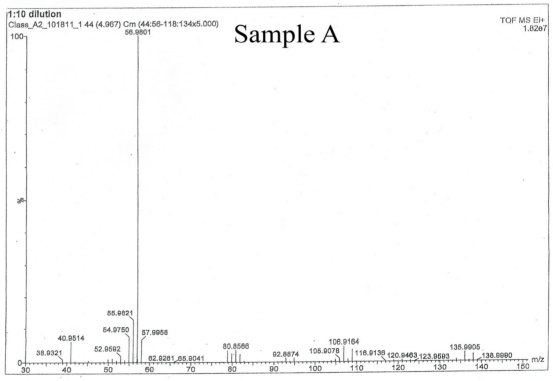
3. Mass spectrometry is used to identify the mass to charge ratio of one or more molecules (broad). A mass spectrometer ionizes the sample by electron impact and then sorts the ions with a magnetic sector mass analyzer. The M+ peak on a GCMS spectra tells the mass of the overall molecule. Isotope abundance can also be identified based on GCMS and probability calculations (4).

Data:

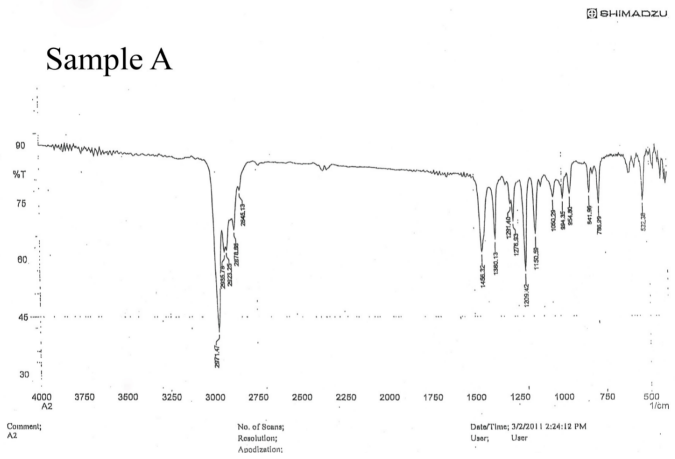
NMR Sample A



GCMS Sample A



FTIR Sample A



Analysis:

1. NMR Identification

   The four signals on the spectrum allow one to infer that there will be a four carbon chain in the molecule. The first peak of the NMR spectrum shows 3 protons and 2 neighbors at 1 ppm . The second peak on the spectrum shows 3 protons and 1 neighbor at 1.7 ppm. The third peak shows 2 protons and four neighbors at 1.9 ppm. The last peak shows one proton and 5 neighbors at 4.1 ppm. These things can lead one to know that the molecule contains one CH, one CH2, and two CH3 groups.

2. FTIR Identification

   Based on the lack of peaks past the 3000 mark on the graph, it can be assumed that there are no functional groups present in this molecule. The presence of a peak at 532.38 tells that there is a bromine present in the compound.

3. GCMS Identification

   For sample A, the M+ peak was at 138.9990 amu. This number gives a good idea of the mass of the molecule. The large peak at 56.9801 shows the loss of a butyl. Bromine was once again observed at the two peaks about 136 and 138. These two peaks represent isotopes of bromine. Br has a molar mass of about 80g/mol. Based on NMR findings, we can conclude that there are 4 carbons, which would have a total molar mass of about 48g/mol and 9 hydrogens which would have a molar mass of 9g/mol total. When added together the overall molar mass of the molecule would be about 137 g/mol. This is consistent with the M+ peak.

Conclusion:

Based on the data from sample A, the molecule can be identified as 2-Bromobutane. NMR showed the number of carbons and hydrogens, FTIR was used to detect the presence of bromine and GCMS was used to confirm the molecule based on molar mass. The formula for this molecule is C4H9Br.

References

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